# An LP Algorithm for Counting Eulerian Orientations Through the Lens of Promise Polymorphisms

Jincheng Guan
guanjincheng@mail.ustc.edu.cn

Shuai Shao
shao10@ustc.edu.cn

Ke Shi
self.ke.shi@gmail.com

School of Computer Science and Technology & Hefei National Laboratory
University of Science and Technology of China

**Abstract**

The weighted Eulerian orientation counting problem (#EO) plays a key role in the complexity classification program for Holant problems. A recent result [MWX25a] established an $\mathrm{FP}^{\mathrm{NP}}$ versus #P-hard dichotomy for #EO problems. On the tractable side of this dichotomy, the supports of all functions admit the ternary XOR operation as a promise polymorphism relaxed in the same direction. It was left open whether all cases on the $\mathrm{FP}^{\mathrm{NP}}$ side are in fact in FP. In this paper, we settle this question by giving a polynomial-time algorithm for all cases on the $\mathrm{FP}^{\mathrm{NP}}$ side. Consequently, we obtain a complete FP versus #P-hard dichotomy for counting weighted Eulerian orientations, and further for complex-valued Holant problems with a nonzero signature of odd arity.

Our algorithm is based on a linear programming relaxation, but we use it in a nonstandard way. Instead of proving that the relaxation is integral and solving the problem directly from an optimal LP solution, we use the relaxation as a structural tool to lift the promise polymorphism condition to an ordinary polymorphism condition. This reveals an affine local structure of the constraint functions, which leads to tractability. A key technical innovation of our paper is to exploit the one-sided promise polymorphism through cubic forms of LP constraints, which interact precisely with the desired ternary XOR polymorphism.

# 1 Introduction

Given an undirected graph $G$, an *Eulerian orientation* of $G$ is an assignment of a direction to every edge such that, at each vertex, the number of incoming edges equals the number of outgoing edges. A graph admits an Eulerian orientation if and only if every vertex has even degree, by the classical Euler theorem. Thus, deciding whether a graph admits an Eulerian orientation is straightforward. The corresponding counting problem is substantially more difficult. Mihail and Winkler showed that counting the Eulerian orientations of a given graph is #P-complete [MW96]. However, the problem may become tractable when suitable local restrictions are imposed on the orientations of edges. An intriguing example arises from the partition function of the six-vertex model [Pau35, Sla41, Rys63], one of the most extensively studied models in statistical physics. In this example, the underlying graph is 4-regular, and at each vertex one incident edge is restricted to be oriented toward the vertex. Counting Eulerian orientations satisfying these local restrictions can be done in polynomial time [CFX18]. More generally, this type of restriction on the orientations of edges can be represented by constraint functions placed at the vertices.

Motivated by the question of which local constraint functions give rise to tractable counting problems, the framework of counting weighted Eulerian orientations, denoted by #EO, was introduced in [CFS20]. In a #EO instance, each vertex $v$ is assigned a constraint function $f_v$, also called a *signature*, whose value determines the local contribution at $v$ given an orientation. Since each edge has two ends, an orientation of an edge is denoted by assigning 0 to the head and 1 to the tail. Thus, for an edge $e = (u, v)$, its two possible orientations are represented by $(0, 1)$ or $(1, 0)$, and locally at every vertex $v$, a 0 represents an incoming edge and a 1 represents an outgoing edge. An Eulerian orientation corresponds to an assignment to each edge (01 or 10) where the numbers of 0's and 1's at each $v$ are equal. Then, in a #EO instance, the signatures are supported on inputs of Hamming weight exactly half of their arity. A complexity dichotomy for this framework was first obtained under a physical symmetry known as *arrow reversal symmetry* (ARS) [CFS20]. Such a dichotomy classifies every problem in the framework as either polynomial-time solvable (tractable) or #P-hard, leaving no intermediate cases within the classification. In this paper, we consider #EO problems defined by arbitrary complex-valued signatures, without assuming ARS. Building on [MWX25a], our main result is a complete FP versus #P-hard dichotomy for this general setting.

The significance of the #EO framework goes beyond the problem of counting Eulerian orientations itself. It plays a fundamental role in the complexity classification of counting problems, as it provides a bridge between counting constraint satisfaction problems (#CSP) [CLX14] and Holant problems [CLX09]. On the one hand, although #EO signatures are required to be supported on half-weight inputs, the #EO framework is expressive enough to encode arbitrary Boolean #CSP, including those whose constraint functions are not supported on half-weight inputs. On the other hand, #EO constitutes an indispensable component in the complexity classification of Holant problems. Holant is a substantially more expressive counting framework that encompasses #CSP, counting perfect matchings [CLX09], partition functions of statistical-physics models such as the six-vertex and eight-vertex models [CFX18, CF23], and the classical simulation of quantum circuits [CGW18]. The dichotomy for real-valued Holant problems relies crucially on the classification of #EO under ARS [SC20]. Under a suitable holographic transformation, ARS signatures correspond precisely to real-valued signatures. More recently, complex-valued Holant problems containing a nonzero signature of odd arity were classified as either solvable in $\mathrm{FP}^{\mathrm{NP}}$ (i.e., in polynomial time with access to an NP oracle), or #P-hard [MWXZ25]. This classification was based on a corresponding $\mathrm{FP}^{\mathrm{NP}}$ versus #P-hard classification for complex-valued #EO problems without assuming ARS [MWX25a]. Consequently, obtaining a complete FP versus #P-hard dichotomy for these Holant problems requires first resolving the remaining $\mathrm{FP}^{\mathrm{NP}}$ cases for #EO problems. The dichotomy established in

this paper provides exactly this missing step.

A major difficulty in classifying general #EO problems is that, once ARS is removed, previously unknown and sometimes rather unexpected tractable classes begin to emerge. The first such signatures appeared in the complexity classification of the six-vertex model [CFX18]. Because these signatures have arity four, they admit relatively explicit descriptions. However, the structure becomes considerably more complicated with higher arities. A recent line of work [MWX25b, ST26] generalized these arity-four examples and discovered new tractable classes of signatures for #EO problems. Remarkably, the base level of the inductively defined classes discovered in [ST26] is characterized precisely by Hadamard codes. This establishes an unexpected connection between two classical mathematical concepts, Eulerian orientations and Hadamard codes, through counting complexity. The appearance of such nontrivial tractable classes illustrates why a complete dichotomy of #EO problems is challenging. Without an exhaustive classification, it is difficult to predict whether further exceptional tractable families remain undiscovered.

The final missing family emerged in [MWX25a], denoted as $\forall 3\mathfrak{t}_{\mathfrak{h}}$ and $\forall 3\mathfrak{f}^{\mathfrak{r}}$ signatures. It is proved in [MWX25a] that apart from these signatures, the #EO problem is #P-hard for all other complex-valued signatures. However, the #EO problems in which all signatures are $\forall 3\mathfrak{t}_{\mathfrak{h}}$, or all are $\forall 3\mathfrak{f}^{\mathfrak{r}}$, and which satisfy a further condition (Theorem 2.17) exhibit a phenomenon that had not previously appeared in the complexity classification of counting problems: they can be solved in polynomial time with access to an NP oracle for the corresponding decision problems, which ask whether there exists an assignment with nonzero evaluation. This yields an $\mathrm{FP}^{\mathrm{NP}}$ versus #P-hard classification for general #EO. If one believes that $\#\mathrm{P} \not\subseteq \mathrm{FP}^{\mathrm{NP}}$, and that the #EO framework does not contain natural #P-intermediate problems, then these exceptional problems should be expected to lie in FP. This was known in part: [MWX25b] gives polynomial-time algorithms for these problems when all signatures are 0-rebalancing, or all are 1-rebalancing (a recursive condition satisfied, for example, by all pure signatures). In general, however, proving this appears to be highly nontrivial, and it remained unresolved in [MWX25a].

To convey how unusual the remaining tractable family is, we review a nontrivial example, $f_{56}$, among the $\forall 3\mathfrak{t}_{\mathfrak{h}}$ and $\forall 3\mathfrak{f}^{\mathfrak{r}}$ signatures. The signature $f_{56}$, introduced in [MWX25b] as a signature that is not rebalancing, has arity 56; whether $\#\mathrm{EO}(f_{56})$ is in FP was left open there. It has a support set $\{\alpha^1, \alpha^2, \ldots, \alpha^5\}$ of size 5, with $f_{56}(\alpha^i) = 1$ for $i \in \{1, \ldots, 5\}$ and $f_{56}(\alpha) = 0$ otherwise. Clearly, $f_{56}$ is completely determined by its support set. Below, we list its five support inputs. Following [MWX25b], define the matrices

$$H_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad H_2 = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}, \quad H_4 = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix}.$$

Also, define a 5-by-56 matrix $R_{56} := [\, H_0 \; H_2^{\to 4} \; H_4^{\to 3} \,]$, where $A^{\to k}$ denotes the concatenation of $k$ copies of $A$. Then, each row of $R_{56}$ is a support input $\alpha^i$ of $f_{56}$. The signature $f_{56}$ has the following properties. Each row $\alpha^i$ has Hamming weight 28: it contains four 1's in each of the four $H_2$ blocks and four 1's in each of the three $H_4$ blocks. Hence $\mathrm{wt}(\alpha^i) = 28$ for every $i \in [5]$, so $f_{56}$ is an EO signature of arity 56. For any pairwise distinct $i, j, k \in [5]$, we have $\mathrm{wt}(\alpha^i \oplus \alpha^j \oplus \alpha^k) = 30 > 28$.

At first sight, there is little reason to expect the counting problem defined by such a signature $f_{56}$ to admit a polynomial-time algorithm. This example illustrates one of the main values of a complexity dichotomy: an exhaustive classification can reveal tractable problems whose algebraic structure and algorithmic tractability would otherwise be very difficult to anticipate.

In this paper, we handle the #EO problems defined by the signature $f_{56}$, as well as all other cases on the $\mathrm{FP}^{\mathrm{NP}}$ side, by giving a unified polynomial-time algorithm. This settles the tractability of all cases previously placed in the $\mathrm{FP}^{\mathrm{NP}}$ class.

**Theorem 1.1.** *Every* #EO *problem belonging to the* $\mathrm{FP}^{\mathrm{NP}}$ *class of the classification in [MWX25a] is solvable in polynomial time. Consequently, the previous* $\mathrm{FP}^{\mathrm{NP}}$ *versus* #P*-hard classification of #EO is strengthened to a complete FP versus* #P*-hard dichotomy.*

As remarked after the main theorem of [MWXZ25], an FP versus #P-hard dichotomy for #EO turns the classification there into one as well. Hence Theorem 1.1 yields the following consequence.

**Corollary 1.2.** *The previous* $\mathrm{FP}^{\mathrm{NP}}$ *versus* #P*-hard classification of complex-valued Holant problems containing a nonzero signature of odd arity is strengthened to a complete FP versus* #P*-hard dichotomy.*

Our algorithm is based on a linear-programming relaxation. We first give an overview of the main ideas. We express the $\forall 3\!\!\uparrow\!\!\rightarrow$ and $\forall 3\!\!\uparrow\!\!\nearrow$ signatures of [MWX25a] by a *promise polymorphism* condition associated with the ternary XOR operation. Our goal is to show that, for a #EO instance whose signatures are all $\forall 3\!\!\uparrow\!\!\rightarrow$ or all $\forall 3\!\!\uparrow\!\!\nearrow$, this promise polymorphism condition can be *efficiently* strengthened to an ordinary polymorphism condition on the local inputs that the relaxation allows. A nonempty set closed under $\oplus_3$ has a much more rigid algebraic structure: it is precisely an affine linear subspace, which eventually leads to tractability. For this purpose, we formulate a linear system using indicator variables for local assignments in truth-table form, rather than the original edge variables of the #EO instance.

Our use of the linear system, however, differs from its more standard role in linear-programming algorithms. We do not solve the linear-programming relaxation directly and output an integral point as a solution to the decision problem corresponding to the counting instance. Recall that this is why an NP oracle is needed in [MWX25a]. In fact, the relaxation is not integral, even when all supports are affine. Rather, the linear-programming relaxation serves as a structural tool for lifting the promise polymorphism condition to an ordinary polymorphism condition, in which the key is a balance identity satisfied by Eulerian orientation instances. In addition, our analysis of the linear system is uncommon and may initially appear unnatural. Instead of analyzing the linear system itself, we study its cubic expressions, which interact precisely with the ternary XOR operation underlying the promise polymorphism condition. Concretely, applying $\oplus_3$ to three independent samples from an LP solution gives a new distribution at each vertex, its threefold convolution, and the cubic expressions are its edge constraints; showing that it is again an LP solution proves that the LP-feasible local inputs are closed under $\oplus_3$ (Section 4).

Thus, the main technical contribution of this paper is not merely the removal of an NP oracle. It also provides a structural explanation for why the exceptional $\mathrm{FP}^{\mathrm{NP}}$ cases are tractable: the constraints imposed by a #EO instance eliminate the freedom allowed by the promise even in the linear-programming relaxation, and force the promise polymorphism condition to become a tractable polymorphism condition.

**Relation to promise CSPs** Promise polymorphisms, originally called weak polymorphisms, were introduced by Austrin, Guruswami, and Håstad [AGH17] and subsequently developed in the algebraic theory of promise constraint satisfaction problems (PCSPs) [BG21, BBKO21]. They have led to complexity classifications for symmetric Boolean PCSPs and algorithms combining Linear programming with algebraic methods [BG21, BG19, BGWŽ20]. In particular, the idea of

combining linear programming and affine relaxation was used in [BGWŽ20] to remove infeasible local assignments.

In this paper, we explore the idea of LP and affine relaxations together with support structures specific to #EO, half-weighted supports and a uniform one-sided relaxation of the ternary XOR polymorphism. A key technical innovation of our paper is to exploit the one-sided promise polymorphism through cubic forms of the LP constraints. Under the $\{\pm 1\}$ encoding, ternary XOR becomes coordinatewise multiplication, so the cubic expansion records the Hamming-weight imbalance of XOR outputs. The one-sided condition makes all these imbalance terms have the same sign, while the Eulerian orientation constraints force their sum to vanish. This interaction between the promise condition and the structure of #EO is the key ingredient beyond the standard use of promise polymorphisms in PCSPs.

# 2 Preliminaries

## 2.1 Counting Eulerian orientations

A signature of arity $r$ is a function $f : \{0,1\}^r \to \mathbb{C}$. We write $\mathrm{Var}(f) = (x_1, \ldots, x_r)$ for the variables of $f$ in a fixed order. For an input $\alpha = (\alpha_1, \ldots, \alpha_r) \in \{0,1\}^r$, the value of $f$ on $\alpha$ is denoted by $f(\alpha)$. We can also view $\alpha$ as a 0-1 bit-string. In this notation, the $i$-th bit $\alpha_i$ of the string is the value assigned to the $i$-th variable $x_i$ of $f$, which can also be written as $\alpha(x_i)$. Note that the bit-string notation is always understood with respect to the given order of the variables.

**Definition 2.1 (Support set, $\mathcal{S}(f)$).** *The support set of a signature $f$ of arity $r$, denoted by $\mathcal{S}(f)$, is the set of all input strings on which $f$ takes non-zero values, i.e., $\mathcal{S}(f) = \{\alpha \in \{0,1\}^r \mid f(\alpha) \neq 0\}$.*

For a positive integer $n$, let $[n] := \{1, 2, \ldots, n\}$. Suppose that $|\mathcal{S}(f)| = m$. Fix an order of all strings in $\mathcal{S}(f) = \{\alpha^1, \alpha^2, \ldots, \alpha^m\}$. We give the following truth-table representation $T(f)$ of $f$ which can be written as an $m \times r$ matrix where the $i$-th row is the support $\alpha^i$, and the columns are indexed by the variables $\mathrm{Var}(f) = (x_1, x_2, \ldots, x_r)$ of $f$ in a given order. Then, $T_{i,j}(f) = \alpha^i(x_j)$.

We define the following useful sets. For a 0-1 bit string $\alpha$ of even length $2d$, let $|\alpha|$ denote its length and let $\mathrm{wt}(\alpha)$ denote its Hamming weight, i.e., the number of 1's in $\alpha$. Define $\mathrm{HW}^{=} := \{\alpha \in \{0,1\}^* \mid \mathrm{wt}(\alpha) = \frac{1}{2}|\alpha|\}, \mathrm{HW}^{>} := \{\alpha \in \{0,1\}^* \mid \mathrm{wt}(\alpha) > \frac{1}{2}|\alpha|\}$, and $\mathrm{HW}^{<} := \{\alpha \in \{0,1\}^* \mid \mathrm{wt}(\alpha) < \frac{1}{2}|\alpha|\}$. A signature $f$ is called an EO signature if $\mathcal{S}(f) \subseteq \mathrm{HW}^{=}$.

For an undirected graph $G = (V, E)$, an orientation $\sigma$ of $G$ is an assignment of a direction to each edge of $G$. Under our convention, a direction $u \to v$ of an edge $e = (u, v)$ can be encoded by assigning the value 1 to the endpoint $u$ and the value 0 to the endpoint $v$. Moreover, an orientation $\sigma$ is called an Eulerian orientation if the number of incoming edges equals the number of outgoing edges at every vertex, or equivalently, every vertex takes the same number of 0 inputs and 1 inputs. We denote by $\mathrm{EO}(G)$ the set of all Eulerian orientations of $G$.

**Definition 2.2 (Counting EO).** *A counting complex-weighted Eulerian orientation problem is parametrized by a set of EO signatures $\mathcal{F}$, and is denoted by $\#\mathrm{EO}(\mathcal{F})$. An instance of $\#\mathrm{EO}(\mathcal{F})$ is a signature grid $\Omega = (G, \pi)$ over $\mathcal{F}$, which consists of a graph $G = (V, E)$, possibly with parallel edges and self-loops, and a mapping $\pi$ that assigns to each vertex $v \in V$ an $f_v \in \mathcal{F}$ and a linear order of the incident edges at $v$. The output is*

$$\#\mathrm{EO}_\Omega = \sum_{\sigma \in \mathrm{EO}(G)} \prod_{v \in V} f_v(\sigma|_{E(v)}),$$

*where $\sigma$ is an Eulerian orientation of $G$ and $\sigma|_{E(v)}$ is the orientation $\sigma$ restricted to the set $E(v)$ of incident edges of $v$.*

**Definition 2.3 (Effective orientations and decision EO).** *Given a* #EO *instance $\Omega = (G, \pi)$ where $G = (V, E)$ and $\pi(v) = f_v$, we say $\sigma \in \mathrm{EO}(G)$ is an effective Eulerian orientation of $\Omega$ if $\prod_{v \in V} f_v(\sigma|_{E(v)}) \neq 0$. The Decision* EO *problem is to decide whether there exists an effective Eulerian orientation $\sigma \in \mathrm{EO}(G)$ for an instance $\Omega = (G, \pi)$.*

**Definition 2.4 (Locally effective support set, $\mathcal{E}(f_v)$).** *Given a* #EO *instance $\Omega = (G, \pi)$ where $G = (V, E)$ and $\pi(v) = f_v$, we say an input string $\alpha \in \mathcal{S}(f_v)$ is locally effective for $f_v$ if there exists an effective Eulerian orientation $\sigma \in \mathrm{EO}(G)$ of $\Omega$ such that $\alpha = \sigma|_{E(v)}$. We use $\mathcal{E}(f_v)$ to denote the set of all locally effective strings for $f_v$, namely the* locally effective support set.

For a signature $f$ of arity $r$ and a set $\mathcal{T} \subseteq \{0,1\}^r$, we write $f|_{\mathcal{T}}$ for the signature obtained by restricting the support set of $f$ to $\mathcal{T}$, namely $(f|_{\mathcal{T}})(\alpha) = f(\alpha)$ if $\alpha \in \mathcal{T}$ and $(f|_{\mathcal{T}})(\alpha) = 0$ otherwise.

**Lemma 2.5.** *Let $\Omega = (G, \pi)$ with $G = (V, E)$ and $\pi(v) = f_v$ be a* #EO *instance. For every $v \in V$, suppose that $\mathcal{T}_v$ is a set satisfying $\mathcal{E}(f_v) \subseteq \mathcal{T}_v$. Consider another* #EO *instance $\Omega' = (G, \pi')$ on the same underlying graph $G$ where $\pi'(v) = f_v\,|_{\mathcal{T}_v}$. Then, we have $\#\mathrm{EO}_\Omega = \#\mathrm{EO}_{\Omega'}$.*

**Proof.** By the definition of Counting EO, $\#\mathrm{EO}_\Omega = \sum_{\sigma \in \mathrm{EO}(G)} \prod_{v \in V} f_v(\sigma|_{E(v)})$. We compare the two sums term by term. If $\sigma$ is an effective Eulerian orientation of $\Omega$, then for every $v \in V$, the string $\sigma|_{E(v)}$ is locally effective, since $\mathcal{E}(f_v) \subseteq \mathcal{T}_v$, we get $(f_v|_{\mathcal{T}_v})(\sigma|_{E(v)}) = (f_v|_{\mathcal{E}(f_v)})(\sigma|_{E(v)})$ for every $v \in V$, so the term of $\sigma$ is unchanged. If $\sigma$ is not an effective Eulerian orientation, then $\prod_{v \in V} f_v(\sigma|_{E(v)}) = 0$, thus $\prod_{v \in V}(f_v|_{\mathcal{E}(f_v)})(\sigma|_{E(v)}) = 0$ and $\prod_{v \in V}(f_v|_{\mathcal{T}_v})(\sigma|_{E(v)}) = 0$. Hence $\#\mathrm{EO}_\Omega = \#\mathrm{EO}_{\Omega'}$. □

Below, we give a truth-table representation, namely truth-table indicator vector, for effective Eulerian orientations. We will define a linear system in terms of these vectors.

**Definition 2.6 (Truth-table indicator vector).** *Let $\Omega = (G, \pi)$ be a* #EO *instance where $G = (V, E)$ and $\pi(v) = f_v$ for every $v \in V$. Suppose that $|\mathcal{S}(f_v)| = m_v$, and $\mathcal{S}(f_v)$ is listed in a fixed order. For every effective Eulerian orientation $\sigma$ of $\Omega$, we define the following 0-1 valued vector $\boldsymbol{\lambda}^\sigma = (\lambda^\sigma_{v,\alpha})_{\alpha \in \mathcal{S}(f_v),\, v \in V} \in \{0,1\}^{\sum_{v \in V} m_v}$, where $\lambda^\sigma_{v,\alpha} = 1$ if $\sigma|_{E(v)} = \alpha$ and $\lambda^\sigma_{v,\alpha} = 0$ otherwise. We say $\boldsymbol{\lambda}^\sigma$ is an effective truth-table indicator of the instance $\Omega$.*

## 2.2 Promise polymorphism

We use $\oplus_3 : \{0,1\}^3 \to \{0,1\}$ to denote the ternary XOR operation, i.e., $\oplus_3(a, b, c) = a \oplus b \oplus c$. The notion of polymorphism [JCG97, BJK05] is used in the algebraic approach to the CSP. It is usually defined for relations, i.e., $\{0,1\}$-valued functions. We extend it to complex-valued signatures by considering its support set. We use its one-sided relaxation, the promise polymorphism of promise CSPs [AGH17, BG18, BBKO21], as a key notion in our linear programming algorithm.

**Definition 2.7 (Polymorphism and promise polymorphism).** *For an* EO *signature $f$ of arity $2d$, we say $f$ admits $\oplus_3$*

1. *as a* polymorphism*, denoted by $f \in \mathsf{Pol}(\oplus_3)$, if for all $\alpha, \beta, \gamma \in \mathcal{S}(f)$, $\alpha \oplus \beta \oplus \gamma \in \mathcal{S}(f)$,*
2. *as an* up promise polymorphism*, denoted by $f \in \mathsf{Pol}^{\uparrow}(\oplus_3)$, if for all $\alpha, \beta, \gamma \in \mathcal{S}(f)$, $\alpha \oplus \beta \oplus \gamma \in \mathcal{S}(f) \cup \mathrm{HW}^{>}$;*

3. *and as a* down promise polymorphism, *denoted by* $f \in \mathsf{Pol}^{\downarrow}(\oplus_3)$, *if for all* $\alpha, \beta, \gamma \in \mathcal{S}(f)$, $\alpha \oplus \beta \oplus \gamma \in \mathcal{S}(f) \cup \mathrm{HW}^{<}$.

*In PCSP terminology [BG18, BBKO21],* $f \in \mathsf{Pol}^{\uparrow}(\oplus_3)$ *means that* $\oplus_3$ *is a polymorphism of the promise template* $(\mathbf{A}, \mathbf{B})$, *where* $\mathbf{A}$ *has the relations* $\mathcal{S}(f)$ *and* $\{01, 10\}$, *and* $\mathbf{B}$ *has the relations* $\mathcal{S}(f) \cup \mathrm{HW}^{>}$ *and* $\{01, 10\}$ *(the disequality models the edges, and* $\oplus_3$ *preserves it);* $f \in \mathsf{Pol}^{\downarrow}(\oplus_3)$ *means the same with* $\mathrm{HW}^{<}$ *in place of* $\mathrm{HW}^{>}$.

*For a signature set* $\mathcal{F}$, *if for every* $f \in \mathcal{F}$, *we have* $f \in \mathsf{Pol}^{\uparrow}(\oplus_3)$, *then we say* $\mathcal{F}$ *admits* $\oplus_3$ *as an up promise polymorphism, denoted by* $\mathcal{F} \subseteq \mathsf{Pol}^{\uparrow}(\oplus_3)$. *Symmetrically, we can define* $\mathcal{F} \subseteq \mathsf{Pol}^{\downarrow}(\oplus_3)$. *Similarly, we can define* $\mathcal{F} \subseteq \mathsf{Pol}(\oplus_3)$.

**Remark 2.8.** *If a signature* $f$ *admits* $\oplus_3$ *as a polymorphism, then its support set* $\mathcal{S}(f)$ *is an* affine linear subspace *over* $\mathbb{F}_2$. *We simply say* $\mathcal{S}(f)$ *is affine, or* $f$ *has an affine support; by convention, the empty set is also affine.*

*Signatures in* $\mathsf{Pol}^{\uparrow}(\oplus_3)$ *and* $\mathsf{Pol}^{\downarrow}(\oplus_3)$ *were originally introduced in [MWX25a], and denoted as* $\forall 3\!\uparrow\!\searrow$ *and* $\forall 3\!\downarrow\!\nearrow$ *signatures respectively. In this paper, we view these signatures through the lens of promise polymorphisms, which relax the standard definition of a polymorphism in a one-sided way.*

The following lemma is proved in [MWX25a] for #EO instances with signatures admitting $\oplus_3$ as an up or down promise polymorphism.

**Lemma 2.9 ([MWX25a]).** *Let* $\Omega = (G, \pi)$ *with* $G = (V, E)$ *and* $\pi(v) = f_v$ *be an instance of* $\#\mathrm{EO}(\mathcal{F})$. *If* $\mathcal{F} \subseteq \mathsf{Pol}^{\uparrow}(\oplus_3)$ *or* $\mathcal{F} \subseteq \mathsf{Pol}^{\downarrow}(\oplus_3)$, *then for every* $v \in V$, *the locally effective support set* $\mathcal{E}(f_v)$ *is affine.*

## 2.3 Known results

We recall several theorems that will be used in this paper: two classes of polynomial-time solvable EO signatures, $\mathscr{A}$ and $\mathscr{P}$, and the $\mathrm{FP}^{\mathrm{NP}}$ versus #P-hard classification for #EO [MWX25a].

**Definition 2.10 (The classes $\mathscr{A}$ and $\mathscr{P}$).** *We use* $\mathscr{A}$ *and* $\mathscr{P}$ *for the two standard tractable classes in the* #CSP *dichotomy of [CLX14]. A signature* $f : \{0,1\}^r \to \mathbb{C}$ *belongs to* $\mathscr{A}$ *if either* $f \equiv 0$, *or there exist a nonzero constant* $\lambda \in \mathbb{C}$, *a matrix* $A$ *over* $\mathbb{F}_2$, *and* 0-1*-valued affine linear forms* $\ell_1, \ldots, \ell_m$ *over* $\mathbb{F}_2$ *such that, for* $X = (x_1, \ldots, x_r, 1)^T$,

$$f(x_1, \ldots, x_r) = \lambda \, \chi_{AX=0} \, i^{\ell_1(X)+\cdots+\ell_m(X)}.$$

*Here* $\chi$ *is a 0-1 indicator function such that* $\chi_{AX=0} = 1$ *iff* $AX = 0$, *while the exponent* $\ell_1(X) + \cdots + \ell_m(X)$ *is an ordinary integer sum, equivalently read modulo* 4. *The class* $\mathscr{P}$ *consists of signatures expressible as products of unary signatures, binary equality signatures* $=_2$, *and binary disequality signatures* $\neq_2$.

**Remark 2.11.** *If a signature* $f$ *is in* $\mathscr{A}$ *or* $\mathscr{P}$, *then its support* $\mathcal{S}(f)$ *is affine.*

**Theorem 2.12 (Tractable classes $\mathscr{A}$ and $\mathscr{P}$ [CFS20]).** *Let* $\mathcal{F}$ *be a set of* EO *signatures. If* $\mathcal{F} \subseteq \mathscr{A}$ *or* $\mathcal{F} \subseteq \mathscr{P}$, *then* $\#\mathrm{EO}(\mathcal{F}) \in \mathrm{FP}$.

Given $2d$ variables $x_1, x_2, \ldots, x_{2d} \in \{0,1\}$ in a fixed order, a perfect pairing $M$ of them is a partition of the $2d$ variables into $d$ pairs, namely $M = \{\{x_{i_1}, x_{i_2}\}, \{x_{i_3}, x_{i_4}\}, \ldots, \{x_{i_{2d-1}}, x_{i_{2d}}\}\}$. Fixing a perfect pairing $M$ of the $2d$ variables, we define

$$\{0,1\}^M := \{(x_1, x_2, \ldots, x_{2d}) \in \{0,1\}^{2d} \mid x_{i_1} \neq x_{i_2}, x_{i_3} \neq x_{i_4}, \ldots, x_{i_{2d-1}} \neq x_{i_{2d}}\}.$$

**Lemma 2.13 ([CFS20]).** *If an* EO *signature $f$ has an affine support, i.e., $f \in \mathsf{Pol}(\oplus_3)$, then there exists a perfect pairing $M$ of the variables of $f$ such that $\mathcal{S}(f) \subseteq \{0,1\}^M$. Moreover, $M$ can be found in time polynomial in the arity of $f$ and $|\mathcal{S}(f)|$.*

**Definition 2.14 (The classes $\mathrm{EO}^{\mathscr{A}}$ and $\mathrm{EO}^{\mathscr{P}}$ [MWX25b, MWX25a]).** *We say that an* EO *signature $f$ of arity $2d$ is in the class $\mathrm{EO}^{\mathscr{A}}$ if for any perfect pairing $M$ of the $2d$ variables of $f$, we have $f|_{\{0,1\}^M} \in \mathscr{A}$. We say that $f$ is in the class $\mathrm{EO}^{\mathscr{P}}$ if for any perfect pairing $M$ of the $2d$ variables of $f$, $f|_{\{0,1\}^M} \in \mathscr{P}$.*

**Lemma 2.15.** *If $f \in \mathrm{EO}^{\mathscr{P}}$ or $\mathrm{EO}^{\mathscr{A}}$, then for any affine linear subspace $\mathcal{T} \subseteq \mathcal{S}(f)$, there exists some $\mathcal{T}'$ satisfying $\mathcal{T} \subseteq \mathcal{T}'$ such that $f\,|_{\mathcal{T}'} \in \mathscr{P}$ or $\mathscr{A}$ respectively.*

*In particular, if $f \in \mathrm{EO}^{\mathscr{A}}$, then $f\,|_{\mathcal{T}'}$ can be simply replaced by $f\,|_{\mathcal{T}}$.*

**Proof.** Suppose that $f$ is an EO signature of arity $2d$. Since $f|_{\mathcal{T}}$ is an EO signature and has affine support, by Lemma 2.13, there exists a perfect pairing $M$ of $2d$ variables of $f|_{\mathcal{T}}$ such that $\mathcal{T} \subseteq \{0,1\}^M$. Since $f \in \mathrm{EO}^{\mathscr{P}}$ or $\mathrm{EO}^{\mathscr{A}}$, so $f|_{\{0,1\}^M} \in \mathscr{P}$ or $\mathscr{A}$ respectively.

Now we consider the case that $f \in \mathrm{EO}^{\mathscr{A}}$, by the above discussion, we have $f|_{\{0,1\}^M} \in \mathscr{A}$. By the definition of $\mathscr{A}$, suppose that $\mathrm{Var}(f) = \{x_1, x_2, \ldots, x_{2d}\}$, then $f|_{\{0,1\}^M}(x_1, x_2, \ldots, x_{2d}) = \lambda \chi_{AX=0} i^{\ell_1(X)+\cdots+\ell_m(X)}$ where $X = (x_1, \ldots, x_{2d}, 1)$ and $AX = 0$ is an affine linear system over $\mathbb{F}_2$. Since $\mathcal{T}$ is an affine linear subspace over $\mathbb{F}_2$, there exists a matrix $B$ over $\mathbb{F}_2$ such that $\mathcal{T} = \{(x_1, x_2, \ldots, x_{2d}) : BX = 0\}$. Hence, $f|_{\mathcal{T}}(x_1, \ldots, x_{2d}) = f|_{\{0,1\}^M}(x_1, \ldots, x_{2d})\chi_{BX=0} = \lambda \chi_{AX=0}\chi_{BX=0} i^{\ell_1(X)+\cdots+\ell_m(X)} = \lambda \chi_{CX=0} i^{\ell_1(X)+\cdots+\ell_m(X)} \in \mathscr{A}$, where $C = \left[\begin{smallmatrix} A \\ B \end{smallmatrix}\right]$. □

**Remark 2.16.** *We give an example to illustrate why, for $f \in \mathrm{EO}^{\mathscr{P}}$, we cannot directly get $f\,|_{\mathcal{T}} \in \mathscr{P}$ for any affine set $\mathcal{T} \subseteq \mathcal{S}(f)$. Let $f(x_1, y_1, x_2, y_2, x_3, y_3) = \prod_{i=1}^3 u(x_i) \prod_{i=1}^3 \neq_2 (x_i, y_i)$, where $\neq_2 (x_i, y_i) = \chi_{x_i \oplus y_i = 1}$ is the binary disequality function, and $u(x_i)$ is a unary function with $u(0) = 1$ and $u(1) = 3$. Let $\mathcal{T} = \{(x_1, y_1, x_2, y_2, x_3, y_3) : x_1 \oplus y_1 = 1, x_2 \oplus y_2 = 1, x_3 \oplus y_3 = 1, x_1 \oplus x_2 \oplus x_3 = 1\}$, which is clearly an affine linear subspace. One can check that $f \in \mathrm{EO}^{\mathscr{P}}$ and indeed $f \in \mathscr{P}$, but $f|_{\mathcal{T}} \notin \mathscr{P}$.*

**Theorem 2.17 ($\mathrm{FP}^{\mathrm{NP}}$ versus #P-hard classification for #EO [MWX25a]).** *Let $\mathcal{F}$ be a set of* EO *signatures. If $\mathcal{F}$ satisfies the following two conditions:*

1. *$\mathcal{F} \subseteq \mathsf{Pol}^{\uparrow}(\oplus_3)$ or $\mathcal{F} \subseteq \mathsf{Pol}^{\downarrow}(\oplus_3)$,*
2. *$\mathcal{F} \subseteq \mathrm{EO}^{\mathscr{A}}$ or $\mathcal{F} \subseteq \mathrm{EO}^{\mathscr{P}}$,*

*then $\#\mathrm{EO}(\mathcal{F}) \in \mathrm{FP}^{\mathrm{NP}}$. Otherwise, $\#\mathrm{EO}(\mathcal{F})$ is #P-hard.*

We briefly review the proof idea for the $\mathrm{FP}^{\mathrm{NP}}$ side and explain why an NP oracle is needed.

Consider an instance $\Omega = (G, \pi)$ of $\#\mathrm{EO}(\mathcal{F})$, where $G = (V, E)$ and $\pi(v) = f_v \in \mathcal{F}$. Since either $\mathcal{F} \subseteq \mathsf{Pol}^{\uparrow}(\oplus_3)$ or $\mathcal{F} \subseteq \mathsf{Pol}^{\downarrow}(\oplus_3)$, Lemma 2.9 implies that $\mathcal{E}(f_v)$ is affine for every $v \in V$. Moreover, since either $\mathcal{F} \subseteq \mathrm{EO}^{\mathscr{A}}$ or $\mathcal{F} \subseteq \mathrm{EO}^{\mathscr{P}}$, Lemma 2.15 implies that, for every $v \in V$, there exists a set $\mathcal{T}'_v$ satisfying $\mathcal{E}(f_v) \subseteq \mathcal{T}'_v \subseteq \mathcal{S}(f_v)$ such that either $f_v\,|_{\mathcal{T}'_v} \in \mathscr{A}$ for all $v \in V$, or $f_v\,|_{\mathcal{T}'_v} \in \mathscr{P}$ for all $v \in V$.

Now consider the instance $\Omega' = (G, \pi')$, where either $\pi'(v) = f_v\,|_{\mathcal{T}'_v} \in \mathscr{A}$ for all $v \in V$, or $\pi'(v) \in \mathscr{P}$ for all $v \in V$. By Lemma 2.5, we have $\#\mathrm{EO}_{\Omega} = \#\mathrm{EO}_{\Omega'}$. Clearly, the value of $\#\mathrm{EO}_{\Omega'}$ can be computed in polynomial time. However, the difficulty is that computing $\mathcal{E}(f_v)$ is not known to be polynomial-time solvable. In [MWX25a], $\mathcal{E}(f_v)$ is obtained by deciding, for each $\alpha \in \mathcal{S}(f_v)$, whether $\alpha$ is locally effective. This decision problem is in NP, which is why an NP oracle is needed.

## 3 An LP-based algorithm

In this section, we introduce our linear programming (LP)-based algorithm. We show how we use LP as a structural tool for lifting the promise polymorphism condition to an ordinary polymorphism condition. Remarkably, our algorithm does not compute $\mathcal{E}(f_v)$, although $\mathcal{E}(f_v)$ was proved to be affine in [MWX25a]. Instead, we give a polynomial-time algorithm to compute a possibly larger set $\mathcal{L}(f_v)$, called the LP-feasible support set (Definition 3.2) satisfying $\mathcal{E}(f_v) \subseteq \mathcal{L}(f_v) \subseteq \mathcal{S}(f_v)$. We further prove that $\mathcal{L}(f_v)$ is affine by analyzing linear-programming constraints. This gives tractability.

Now, we define a linear system for an instance $\Omega = (G, \pi)$ of $\#\mathrm{EO}(\mathcal{F})$ where $\mathcal{F}$ satisfies the two conditions in Theorem 2.17. We give some notations. Suppose $G = (V, E)$. For every $v \in V$ with degree $2d_v$, let $f_v \in \mathcal{F}$ of arity $2d_v$ be the signature labeled on $v$, i.e., $\pi(v) = f_v$. Let $e \in E$ be an edge incident to $v$ and $\alpha \in \mathcal{S}(f_v)$ be a support vector of $f_v$. We write $\alpha(e)$ for the bit of $\alpha$ at the input of $f_v$ that corresponds to the edge $e$. Here we assume that $G$ has no self-loops: replacing a self-loop at $v$ by two parallel edges to a new vertex labeled $\neq_2 \in \mathsf{Pol}(\oplus_3) \cap \mathrm{EO}^{\mathscr{A}} \cap \mathrm{EO}^{\mathscr{P}}$ does not change $\#\mathrm{EO}_\Omega$.

Now, we give the following linear system.

$$
\begin{aligned}
&\textbf{Variables:} && \lambda_{v,\alpha} \geq 0 && \text{for all } v \in V \text{ and } \alpha \in \mathcal{S}(f_v),\\
&\textbf{Vertex constraints:} && \sum_{\alpha \in \mathcal{S}(f_v)} \lambda_{v,\alpha} = 1 && \text{for every } v \in V,\\
&\textbf{Edge constraints:} && \sum_{\alpha \in \mathcal{S}(f_u)} \alpha(e)\, \lambda_{u,\alpha} + \sum_{\alpha \in \mathcal{S}(f_v)} \alpha(e)\, \lambda_{v,\alpha} = 1 && \text{for every } e = (u, v) \in E.
\end{aligned}
\tag{1}
$$

We use $\boldsymbol{\lambda} = (\lambda_{v,\alpha})_{\alpha \in \mathcal{S}(f_v),\, v \in V}$ to denote the vector of variables, $\mathsf{L}(\Omega)$ to denote the above system, and $\mathsf{P}(\Omega)$ to denote the feasible region of $\mathsf{L}(\Omega)$, which is a polytope. We give the following two key properties of $\mathsf{L}(\Omega)$. First, $\mathsf{P}(\Omega)$ is a linear relaxation of the solution space in terms of truth-table indicator vectors (Definition 2.6) for the corresponding decision EO problem.

**Lemma 3.1.** *If $\sigma$ is an effective Eulerian orientation of $\Omega$, then its effective truth-table indicator $\boldsymbol{\lambda}^\sigma$ (Definition 2.6) lies in $\mathsf{P}(\Omega)$.*

**Proof.** Obviously, $\boldsymbol{\lambda}^\sigma$ is a vector of variables of $\mathsf{L}(\Omega)$ and $\boldsymbol{\lambda}^\sigma \geq 0$. By Definition 2.6, for every vertex $v \in V$, there is exactly one string $\alpha \in \mathcal{S}(f_v)$ such that $\lambda^\sigma_{v,\alpha} = 1$, and $\lambda^\sigma_{v,\beta} = 0$ for all other $\beta \in \mathcal{S}(f_v)$ where $\beta \neq \alpha$. Thus, $\boldsymbol{\lambda}^\sigma$ satisfies the vertex constraints of $\mathsf{L}(\Omega)$.

Since $\sigma$ is an effective Eulerian orientation of $\Omega$, for every edge $e = (u, v) \in E$, $\sigma$ assigns a value 1 to one endpoint and a value 0 to the other endpoint. Then, for every $e = (u, v) \in E$, we have $\sum_{\alpha \in \mathcal{S}(f_u)} \alpha(e)\, \lambda^\sigma_{u,\alpha} + \sum_{\alpha \in \mathcal{S}(f_v)} \alpha(e)\, \lambda^\sigma_{v,\alpha} = \sigma|_{E(v)}(e) + \sigma|_{E(u)}(e) = 1$. Thus, $\boldsymbol{\lambda}^\sigma$ also satisfies the edge constraints of $\mathsf{L}(\Omega)$. Therefore, $\boldsymbol{\lambda}^\sigma \in \mathsf{P}(\Omega)$. □

**Definition 3.2 (LP feasible support set, $\mathcal{L}(f_v)$).** *Given a $\#\mathrm{EO}$ instance $\Omega = (G, \pi)$ where $G = (V, E)$ and $\pi(v) = f_v$, we say an input $\alpha \in \mathcal{S}(f_v)$ is LP feasible for $f_v$ if $\max_{\boldsymbol{\lambda} \in \mathsf{P}(\Omega)} \lambda_{v,\alpha} > 0$. We use $\mathcal{L}(f_v)$ to denote the set of all LP feasible inputs for $f_v$, namely the* LP feasible support set.

**Lemma 3.3.** *Let $\Omega = (G, \pi)$ be a $\#\mathrm{EO}$ instance and $v$ a vertex of $G$. The locally effective support set $\mathcal{E}(f_v)$ (Definition 2.4), the LP feasible support set $\mathcal{L}(f_v)$ (Definition 3.2) and the support set $\mathcal{S}(f_v)$ satisfy $\mathcal{E}(f_v) \subseteq \mathcal{L}(f_v) \subseteq \mathcal{S}(f_v)$.*

**Proof.** By the definition of $\mathcal{E}(f_v)$, for every vertex $v \in V$ and each locally effective string $\beta \in \mathcal{E}(f_v)$, there exists an effective Eulerian orientation $\sigma$ with $\sigma|_{E(v)} = \beta$, and its effective truth-table indicator $\boldsymbol{\lambda}^\sigma$ satisfies $\lambda^\sigma_{v,\beta} = 1 > 0$. And by Lemma 3.1, we have $\boldsymbol{\lambda}^\sigma \in \mathsf{P}(\Omega)$, so that $\max_{\boldsymbol{\lambda}\in\mathsf{P}(\Omega)} \lambda_{v,\beta} \geq \lambda^\sigma_{v,\beta} = 1 > 0$ and we get $\beta \in \mathcal{L}(f_v)$. Then $\mathcal{E}(f_v) \subseteq \mathcal{L}(f_v)$. By the definition of $\mathcal{L}(f_v)$, we have $\mathcal{L}(f_v) \subseteq \mathcal{S}(f_v)$. Finally, $\mathcal{E}(f_v) \subseteq \mathcal{L}(f_v) \subseteq \mathcal{S}(f_v)$. □

Second, for every vertex $v \in V$, the LP feasible support set $\mathcal{L}(f_v)$ determined by $\mathsf{L}(\Omega)$ is an affine linear subspace of $\mathbb{F}_2^{2d_v}$.

**Theorem 3.4 ($\mathcal{L}(f_v)$ is affine).** *Let $\Omega = (G, \pi)$ with $G = (V, E)$ and $\pi(v) = f_v$ be an instance of #EO($\mathcal{F}$). If $\mathcal{F} \subseteq \mathsf{Pol}^{\uparrow}(\oplus_3)$ or $\mathcal{F} \subseteq \mathsf{Pol}^{\downarrow}(\oplus_3)$, then for every $v \in V$, the LP feasible support set $\mathcal{L}(f_v)$ is affine.*

We give a proof of Theorem 3.4 in Section 4. With Theorem 3.4 in hand, we are ready to give our algorithm.

**Algorithm 1. LP-based algorithm.**

Given an instance of #EO($\mathcal{F}$) where $\mathcal{F}$ satisfies the tractability conditions of Theorem 2.17.

---

INPUT: An instance $\Omega = (G, \pi)$ where $\pi(v) = f_v$ for every $v \in V(G)$
OUTPUT: The evaluation $\#\mathrm{EO}_\Omega$

1 **if** $\mathsf{P}(\Omega) = \emptyset$ **then return** 0 ▷ feasibility check
2 **for each** vertex $v \in V(G)$ **do** ▷ find $\mathcal{L}(f_v)$
3  **for each** $\alpha \in \mathcal{S}(f_v)$ **do**
4  Solve the linear programming problem $\max_{\boldsymbol{\lambda}\in\mathsf{P}(\Omega)} \lambda_{v,\alpha}$
5  Determine $\mathcal{L}(f_v)$
6  Determine $\mathcal{T}'_v \supseteq \mathcal{L}(f_v)$ with $f_v|_{\mathcal{T}'_v}$ in $\mathscr{A}$ if $\mathcal{F} \subseteq \mathrm{EO}^{\mathscr{A}}$, else in $\mathscr{P}$
7 Compute the value $\#\mathrm{EO}_{\Omega'}$ where $\Omega' = (G, \pi')$ with $\pi'(v) = f_v|_{\mathcal{T}'_v}$ ▷ tractable instance
8 **return** the value $\#\mathrm{EO}_{\Omega'}$

---

**Input model.** The input of #EO consists of a graph $G = (V, E)$ and the truth table $T(f_v)$ of $f_v$ for every $v \in V$, i.e., the strings in $\mathcal{S}(f_v)$ together with their values. Let $n = |V|$ and $s = \max_{v\in V} |\mathcal{S}(f_v)|$. In particular, for a fixed finite $\mathcal{F}$ as in [MWX25a], $s = O(1)$ is a fixed parameter.

**Proposition 3.5.** *Algorithm 1 output is correct. The algorithm runs in time* $\mathrm{poly}(n, s)$ *where $n$ is the number of vertices, and $s = \max_{v\in V} |\mathcal{S}(f_v)|$.*

*In particular, for a fixed finite $\mathcal{F}$, the algorithm runs in time* $\mathrm{poly}(n)$.

**Proof.** If $\mathsf{P}(\Omega) = \emptyset$, then $\Omega$ has no effective Eulerian orientation by Lemma 3.1, so $\#\mathrm{EO}_\Omega = 0$ and Line 1 is correct. Otherwise, $\mathcal{E}(f_v) \subseteq \mathcal{L}(f_v) \subseteq \mathcal{T}'_v$ for every $v$ by Lemma 3.3 and Line 6, so Lemma 2.5 gives $\#\mathrm{EO}_\Omega = \#\mathrm{EO}_{\Omega'}$, and the algorithm outputs the correct value.

We analyze the time complexity. The linear system $\mathsf{L}(\Omega)$ has at most $n \cdot s$ variables and $|V| + |E|$ equality constraints. Lines 1–5 solve at most $n \cdot s + 1$ linear programs, each in time polynomial in its size, so we can compute all $\mathcal{L}(f_v)$ in time polynomial in $n$, $|E|$ and $s$. Moreover, the set $\mathcal{T}'_v$, given by the perfect pairing in Lemma 2.15, can be found in time polynomial in the arity of $f_v$ and $s$, and the arity of $f_v$ is at most $2|E|$. By Theorem 3.4 and Lemma 2.15, $\Omega'$ is an instance of #EO($\mathscr{A}$) or #EO($\mathscr{P}$), so by Theorem 2.12, we can compute $\#\mathrm{EO}_{\Omega'}$ in polynomial time. Thus, the total running time is polynomial in $n$, $|E|$ and $s$. □

**Proof of Theorem 1.1.** If $\mathcal{F}$ satisfies the two conditions of Theorem 2.17, then Algorithm 1 computes every instance of #EO($\mathcal{F}$) in polynomial time by Proposition 3.5. If at least one condition fails, then #P-hardness follows from Theorem 2.17. □

# 4 From a promise polymorphism to a polymorphism

We prove Theorem 3.4. Throughout this section, $\Omega = (G, \pi)$ is an instance of $\#\mathrm{EO}(\mathcal{F})$ with $G = (V, E)$ and $\pi(v) = f_v \in \mathcal{F}$, and we assume $\mathcal{F} \subseteq \mathsf{Pol}^{\uparrow}(\oplus_3)$; the case $\mathcal{F} \subseteq \mathsf{Pol}^{\downarrow}(\oplus_3)$ is symmetric, with $\mathrm{HW}^{<}$ in place of $\mathrm{HW}^{>}$. The proof combines a routine pushforward of LP solutions along $\oplus_3$ (Lemma 4.1) with a balance identity specific to Eulerian orientations (Lemma 4.2), which makes the promise collapse (Corollary 4.3).

**Sign representation.** For $\alpha \in \{0,1\}^r$, let $\widehat{\alpha}_i = 2\alpha_i - 1 \in \{-1, 1\}$ and $\Delta(\alpha) = \sum_{i=1}^{r} \widehat{\alpha}_i = 2\,\mathrm{wt}(\alpha) - r$. Thus $\Delta(\alpha)$ is zero, positive, or negative according as $\alpha \in \mathrm{HW}^{=}$, $\alpha \in \mathrm{HW}^{>}$, or $\alpha \in \mathrm{HW}^{<}$, and for an orientation $\sigma$, $\Delta(\sigma|_{E(v)})$ is the out-degree minus the in-degree of $v$. One can check that $\widehat{\alpha \oplus \beta \oplus \gamma} = \widehat{\alpha} \cdot \widehat{\beta} \cdot \widehat{\gamma}$ coordinatewise. (Mapping 1 to $+1$ is the opposite of the usual Fourier convention, which makes no difference for $\oplus_3$.)

**Local distributions.** A *local distribution family* $\boldsymbol{\mu}$ assigns to every $v \in V$ a probability distribution $\mu_{v,\cdot}$ on $\{0,1\}^{E(v)}$; unlike a point of $\mathsf{P}(\Omega)$, it need not be supported on $\mathcal{S}(f_v)$. For $e \in E(v)$, let $p_{v,e}(\boldsymbol{\mu}) := \sum_{\zeta \in \{0,1\}^{E(v)}} \widehat{\zeta}(e)\, \mu_{v,\zeta}$ be the bias of the value on $e$ at $v$. We say that $\boldsymbol{\mu}$ *satisfies the edge constraints* if

$$p_{u,e}(\boldsymbol{\mu}) + p_{v,e}(\boldsymbol{\mu}) = 0 \quad \text{for every } e = (u, v) \in E. \tag{2}$$

Since $p_{v,e}(\boldsymbol{\mu}) = 2 \sum_{\zeta \in \{0,1\}^{E(v)}} \zeta(e)\, \mu_{v,\zeta} - 1$, (2) is equivalent to

$$\sum_{\zeta \in \{0,1\}^{E(u)}} \zeta(e)\, \mu_{u,\zeta} + \sum_{\zeta \in \{0,1\}^{E(v)}} \zeta(e)\, \mu_{v,\zeta} = 1 \quad \text{for every } e = (u, v) \in E,$$

which has the same form as the edge constraints of $\mathsf{L}(\Omega)$ in (1).

To show that $\mathcal{L}(f_v)$ is closed under $\oplus_3$, we need, for $\alpha, \beta, \gamma \in \mathcal{L}(f_v)$, an LP solution giving positive weight to $\alpha \oplus \beta \oplus \gamma$. The natural candidate is the XOR of three independent samples from an LP solution $\boldsymbol{\lambda}$, which gives $\alpha \oplus \beta \oplus \gamma$ weight at least $\lambda_{v,\alpha}\lambda_{v,\beta}\lambda_{v,\gamma}$. It is again an LP solution: applying $\oplus_3$ keeps the two ends of every edge opposite on average (Lemma 4.1), and since the total expected out-degree minus in-degree is zero (Lemma 4.2), the strings that only the promise allows, which lie in $\mathrm{HW}^{>}$, receive no weight (Corollary 4.3). Formally, the *threefold convolution* of $\boldsymbol{\lambda} \in \mathsf{P}(\Omega)$ is the local distribution family $\boldsymbol{\mu}$ given by

$$\mu_{v,\zeta} := \sum_{\substack{\alpha,\beta,\gamma \in \mathcal{S}(f_v) \\ \alpha \oplus \beta \oplus \gamma = \zeta}} \lambda_{v,\alpha}\lambda_{v,\beta}\lambda_{v,\gamma} \qquad (v \in V,\ \zeta \in \{0,1\}^{E(v)}).$$

Each $\mu_{v,\cdot}$ is indeed a probability distribution: its entries are nonnegative, and every triple $(\alpha, \beta, \gamma) \in \mathcal{S}(f_v)^3$ contributes to exactly one entry, namely to $\mu_{v,\alpha\oplus\beta\oplus\gamma}$, so by the vertex constraint of $\mathsf{L}(\Omega)$,

$$\sum_{\zeta \in \{0,1\}^{E(v)}} \mu_{v,\zeta} = \sum_{\alpha,\beta,\gamma \in \mathcal{S}(f_v)} \lambda_{v,\alpha}\lambda_{v,\beta}\lambda_{v,\gamma} = \Big( \sum_{\alpha \in \mathcal{S}(f_v)} \lambda_{v,\alpha} \Big)^3 = 1.$$

In other words, $\mu_{v,\cdot}$ is the distribution of $\alpha \oplus \beta \oplus \gamma$ when $\alpha$, $\beta$, $\gamma$ are drawn independently from $\lambda_{v,\cdot}$.

**Lemma 4.1 (Pushforward).** *For every $\boldsymbol{\lambda} \in \mathsf{P}(\Omega)$, its threefold convolution $\boldsymbol{\mu}$ satisfies the edge constraints, and each $\mu_{v,\cdot}$ is supported on $\mathcal{S}(f_v) \cup \mathrm{HW}^{>}$.*

**Proof.** Since $\widehat{\alpha \oplus \beta \oplus \gamma}(e) = \widehat{\alpha}(e)\widehat{\beta}(e)\widehat{\gamma}(e)$,

$$p_{v,e}(\boldsymbol{\mu}) = \sum_{\alpha,\beta,\gamma \in \mathcal{S}(f_v)} \widehat{\alpha}(e)\widehat{\beta}(e)\widehat{\gamma}(e)\, \lambda_{v,\alpha}\lambda_{v,\beta}\lambda_{v,\gamma} = p_{v,e}(\boldsymbol{\lambda})^3.$$

As $p_{u,e}(\boldsymbol{\lambda}) = -p_{v,e}(\boldsymbol{\lambda})$ for every edge $e = (u, v)$, we get $p_{u,e}(\boldsymbol{\mu}) + p_{v,e}(\boldsymbol{\mu}) = p_{u,e}(\boldsymbol{\lambda})^3 + p_{v,e}(\boldsymbol{\lambda})^3 = 0$. Finally, $\mu_{v,\zeta} > 0$ only if $\zeta = \alpha \oplus \beta \oplus \gamma$ for some $\alpha, \beta, \gamma \in \mathcal{S}(f_v)$, and such a $\zeta$ lies in $\mathcal{S}(f_v) \cup \mathrm{HW}^{>}$ since $f_v \in \mathsf{Pol}^{\uparrow}(\oplus_3)$. □

For the Fourier transform of a probability measure, the degree-one coefficient of $\lambda_{v,\cdot}$ at coordinate $e$ is $\sum_{\alpha \in \mathcal{S}(f_v)} \lambda_{v,\alpha}(-1)^{\alpha(e)} = -p_{v,e}(\boldsymbol{\lambda})$. Thus the identity $p_{v,e}(\boldsymbol{\mu}) = p_{v,e}(\boldsymbol{\lambda})^3$ is the degree-one case of the convolution theorem, and cubing preserves the edge constraints (2).

**Lemma 4.2 (Balance).** *Every local distribution family $\boldsymbol{\mu}$ satisfying the edge constraints satisfies $\sum_{v \in V} \sum_{\zeta \in \{0,1\}^{E(v)}} \mu_{v,\zeta}\, \Delta(\zeta) = 0$.*

**Proof.** Since $\Delta(\zeta) = \sum_{e \in E(v)} \widehat{\zeta}(e)$, we have $\sum_{\zeta \in \{0,1\}^{E(v)}} \mu_{v,\zeta}\Delta(\zeta) = \sum_{e \in E(v)} p_{v,e}(\boldsymbol{\mu})$. Summing over $v$ counts every half-edge once, so every edge $e = (u, v)$ contributes $p_{u,e}(\boldsymbol{\mu}) + p_{v,e}(\boldsymbol{\mu}) = 0$ by (2). □

For an orientation $\sigma$, the identity $\sum_{v \in V} \Delta(\sigma|_{E(v)}) = 0$ holds because every edge has one tail and one head. [MWX25a] applied this integral identity to the XOR of three effective orientations, which is again an orientation, to prove Lemma 2.9. Lemma 4.2 is its fractional version, for local distribution families such as LP solutions and their threefold convolutions.

By the remark after (2), the points of $\mathsf{P}(\Omega)$, extended by zeros, are exactly the local distribution families that satisfy the edge constraints and are supported on $\mathcal{S}(f_v)$ at every $v$; we write $\boldsymbol{\mu} \in \mathsf{P}(\Omega)$ for such families.

**Corollary 4.3 (Collapse).** *For every $\boldsymbol{\lambda} \in \mathsf{P}(\Omega)$, its threefold convolution lies in $\mathsf{P}(\Omega)$.*

**Proof.** By Lemma 4.1, the threefold convolution $\boldsymbol{\mu}$ satisfies the edge constraints and is supported on $\mathcal{S}(f_v) \cup \mathrm{HW}^{>}$ at every $v$. There $\Delta(\zeta) \geq 0$, with equality exactly on $\mathcal{S}(f_v) \subseteq \mathrm{HW}^{=}$, so every term $\mu_{v,\zeta}\Delta(\zeta)$ in Lemma 4.2 is nonnegative and hence zero. Thus $\boldsymbol{\mu}$ puts no weight on $\mathrm{HW}^{>}$, that is, $\mu_{v,\cdot}$ is supported on $\mathcal{S}(f_v)$ for every $v$, and $\boldsymbol{\mu} \in \mathsf{P}(\Omega)$. (Only here must all signatures admit $\oplus_3$ in the same direction, so that all terms have the same sign.) □

**Proof of Theorem 3.4.** If $\mathsf{P}(\Omega) = \emptyset$, every $\mathcal{L}(f_v)$ is empty, hence affine. Otherwise, let $v \in V$ and $\alpha, \beta, \gamma \in \mathcal{L}(f_v)$, not necessarily distinct. Take $\boldsymbol{\lambda}^{(1)}, \boldsymbol{\lambda}^{(2)}, \boldsymbol{\lambda}^{(3)} \in \mathsf{P}(\Omega)$ with $\lambda^{(1)}_{v,\alpha}, \lambda^{(2)}_{v,\beta}, \lambda^{(3)}_{v,\gamma} > 0$. By convexity, $\boldsymbol{\lambda}^* := \frac{1}{3}(\boldsymbol{\lambda}^{(1)} + \boldsymbol{\lambda}^{(2)} + \boldsymbol{\lambda}^{(3)}) \in \mathsf{P}(\Omega)$, and $\lambda^*_{v,\alpha}, \lambda^*_{v,\beta}, \lambda^*_{v,\gamma} > 0$. Its threefold convolution $\boldsymbol{\mu}$ lies in $\mathsf{P}(\Omega)$ by Corollary 4.3, and $\mu_{v,\alpha\oplus\beta\oplus\gamma} \geq \lambda^*_{v,\alpha}\lambda^*_{v,\beta}\lambda^*_{v,\gamma} > 0$, so $\alpha \oplus \beta \oplus \gamma \in \mathcal{L}(f_v)$. Finally, $\mathcal{L}(f_v) \neq \emptyset$, and a nonempty subset of $\mathbb{F}_2^{2d_v}$ closed under $\oplus_3$ is an affine subspace, since its translate by any of its elements is closed under addition. □

**Example 4.4 (The signature $f_{56}$).** *Any three distinct strings in $\mathcal{S}(f_{56})$ have an XOR of weight $30 > 28$, so $f_{56} \in \mathsf{Pol}^{\uparrow}(\oplus_3) \setminus \mathsf{Pol}(\oplus_3)$. As in the proof of Theorem 3.4, Corollary 4.3 puts the XOR of any three strings of $\mathcal{L}(f_v)$ into $\mathcal{S}(f_{56})$; hence $|\mathcal{L}(f_v)| \leq 2$, and $\mathcal{L}(f_v)$ is trivially affine. Thus the LP keeps at most two of the five support strings at each vertex, and $f_{56}|_{\mathcal{L}(f_v)}$, the indicator of at most two strings, lies in $\mathscr{A} \cap \mathscr{P}$, so Line 6 of Algorithm 1 can take $\mathcal{T}'_v = \mathcal{L}(f_v)$.*

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## AI disclosure

During the investigation of a special signature $f_{56}$, we used **GPT-5.4** series models as a research aid. Related materials are available at https://github.com/KeShih/EO (updated on June 26, 2026). The main result and its proof were developed and written by the authors. In preparing this version, we used Claude Opus 5.5 to help revise the exposition.

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